Transformer-Based Autoencoder Framework for Nonlinear Hyperspectral Anomaly Detection

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***Abstract*— Recently, the autoencoder (AE) has received significant attention in the hyperspectral anomaly detection task. However, all existing AE-based anomaly detectors operate under the linear mixing model (LMM), which cannot accurately model the nonlinear mixing phenomenon in practical hyperspectral images (HSIs). Moreover, these AE-based detectors rarely consider the spatial information between pixels, which is crucial to obtain accurate results of anomaly detection. To address the above issues, this article proposes a transformer (TR)-based AE framework (TAEF) for nonlinear hyperspectral anomaly detection. Specifically, the proposed AE framework adopts the TR as the encoder so that not only the local spatial information but also the transitive global spatial information can be considered. The extended multilinear mixing model (EMLM) is embedded into the decoder to accurately characterize the high-order nonlinear mixing phenomenon. By using this TAEF, the background of HSIs can be reconstructed effectively. Moreover, a novel method for generating patches is proposed in this article to support the TR in the characterization of the transitive global spatial information. Besides, to further improve the accuracy of the background reconstruction, the local-clustering method is adopted to decrease the potential anomalies and increase the sparse backgrounds in the meantime. Finally, the anomalous level of pixel is calculated by the reconstruction error. The experimental results on various real hyperspectral datasets demonstrate that the proposed TAEF outperforms the current state-of-the-art (SOTA) anomaly detectors. In addition, our code is available at: https://github.com/I3ab/TAEF.**

***Index Terms*— Autoencoder (AE), extended multilinear mixing model (EMLM), hyperspectral images (HSIs), local clustering, nonlinear anomaly detection, overlapped patches, transformer (TR).**

I. Introduction

HYPERSPECTRAL images (HSIs) with hundreds of spectral bands are collected by sensors on spaceborne or airborne platforms. The HSIs with one spectral dimension and two spatial dimensions can be regarded as 3-D data cubes. Owing to the high spectral resolution of HSIs, each pixel can be viewed as a consecutive spectral curve [1], and each material has its own unique spectral curve, which develops many practical applications in HSIs, such as classification [2], [3] and target detection [4], [5].

Due to that the different spectral curves correspond to the different materials, the HSIs are advantageous in distinguishing and positioning specific targets, which is referred to as target detection. The hyperspectral target detection task can also be known as unsupervised anomaly detection when the target spectra are unknown in advance. Anomaly detection is used to locate unusual objects with particularly small proportions and great differences from their neighbors in spectral characteristics. Given the challenging nature of acquiring the specific target spectra and the advantage of being free from the effects of spectral variability, anomaly detection should be more feasible in real applications, such as search and rescue [6], environmental surveillance [7], and military reconnaissance [8].

Many hyperspectral anomaly detectors based on linear mixing model (LMM) [9] have been proposed in recent decades. One such anomaly detector is the Reed–Xiaoli (RX) [10], which operates on the assumption that the background component of HSIs follows a multivariate Gaussian distribution. Therefore, the anomalous level of each pixel in the RX detector can be expressed by the Mahalanobis distance between the tested pixel and the mean of the background component. However, the RX detector always produces unsatisfactory detection results because the fundamental assumption in RX is hard to be satisfied in practical cases. To avoid making any inaccurate assumptions on the HSIs, the collaborative representation-based detector (CRD) [11] detects anomalies based on the concept that the background pixel can be linearly represented by its adjacent pixels, whereas the anomalous pixel cannot. However, the anomaly detection results of CRD may be significantly influenced by the size of the dual window in different datasets. Besides, the common drawback of both RX and CRD is that each pixel is detected separately and the spatial information in HSIs is ignored, which is essential to obtain accurate results for anomaly detection.

Recently, the importance of spatial information in HSIs has gained increasing attention in many anomaly detectors. The LRaSMD [12] decomposes the HSIs into three components, i.e., anomaly, background, and noise, by incorporating the low-rank and sparse constraints on the background and anomaly, respectively. To further improve the decomposition result of HSIs, many anomaly detectors generate background

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dictionaries using various methods [13], [14] and then spatially constrain the corresponding abundance matrix. Based on this, the GTVLRR incorporates the graph and total variation (TV) [15] regularizations into the low-rank representation (LRR) on the background abundance, which considers the fact that the abundances of the adjacent pixels are also close. Similarly, the LSCTV [16] incorporates the TV regularization in each superpixel block of HSIs to make the background abundances in each superpixel block smooth. However, the LMM-based anomaly detectors mentioned above frequently suffer from significant model errors [17] because of the widespread nonlinear mixing phenomenon caused by complex ground environments, and therefore, the nonlinear mixing models should be considered in hyperspectral anomaly detection.

The kernel trick is the most common technique to address the prevalent nonlinear mixing phenomenon in HSIs due to its ability in mapping the linearly inseparable data into a high-dimensional space, where the mapped data become linearly separable. After feasible kernel mapping, the anomalies are farther away from the background pixels in the high-dimensional space, and thus, the kernel CRD (KCRD) [11] realizes the nonlinear anomaly detection by applying the CRD in the mapped high-dimensional space. Similarly, the anomaly detector with kernel isolation forest (KIFD) [18] also characterizes the separability between anomalies and background pixels by utilizing the isolation forest trick in the high-dimensional space. Nevertheless, all kernel mapping-based methods fail to precisely characterize the inherent nonlinear mixing characteristics between endmembers in HSIs due to the separate mapping of each individual pixel. Furthermore, the procedure for choosing an optimal kernel mapping function is ambiguous, which results in unstable anomaly detection results in these detectors. To solve the above problems, the endmember-kernel theory is proposed in KHype [19] to characterize the nonlinear interactions between endmembers. Based on this theory, the KNUD [20] realizes anomaly detection in the dual window around each tested pixel. All of the above are model-based anomaly detectors, whose major virtue lies in their physical interpretability. However, the disadvantages are also conspicuous that the time cost of these model-based anomaly detectors is relatively high. Besides, the detection results of the above model-based detectors, including linear-based and nonlinear-based, are unsatisfactory because it is difficult to accurately describe various actual HSIs through a common model.

In recent years, the neural network (NN), especially the autoencoder (AE) [21], is increasingly used in hyperspectral anomaly detection tasks because of its exceptional ability in extracting features. To better utilize these extracted features, the LREN [22] integrates the LRR in the latent layer of the AE. Besides, a more common approach is to map the pixels to the hidden layer like the unmixing process and then reconstruct the background because the overwhelming majority of pixels in the HSIs belong to the background and only a few pixels are anomalous. The training target of AE is to minimize the total reconstruction errors of the entire HSIs, and thus, the AE tends to prioritize the accurate reconstruction of the background pixels while disregarding the reconstruction of anomalies. Therefore, the reconstruction error of the well-trained network can be directly used for anomaly detection. On this basis, the RGAE [23] imposes the graph regularization on the latent layer of AE to consider the relationship between pixels. Besides, the Auto-AD [24] builds the AE by the fully convolutional layers to enhance the reconstruction accuracy of background. However, the above AE-based method is likely to extract features mixed with anomalous information, which results in the reduction of the accuracy of background reconstruction and thus the accuracy of anomaly detection. To address this issue, BASGAN [25] applies a coarse search before the training procedure, which considerably reduces the anomalies in the training set. In terms of the same purpose, a guided map is generated to restrain the participation of anomalies in GAED [26]. In contrast to the model-based anomaly detectors, these AE-based anomaly detectors are suitable for a wide range of HSIs with significantly reduced test times. However, the false alarms are serious in these AE-based anomaly detectors due to the relatively small proportion of the sparse backgrounds. Besides, these AE-based anomaly detectors lack sufficient physical interpretability and do not account for nonlinear mixing effects.

To sum up, the above hyperspectral anomaly detectors still suffer from the following problems.

1) For the model-based methods, they have good physical interpretability by using specific spectral mixture models. However, it is difficult for these general models to accurately characterize and describe various complex real HSIs in practical applications, so their detection accuracy is usually unsatisfactory. Moreover, due to the use of iterative recursive optimization algorithms, their detection time is relatively long.
2) For the data-driven methods, although they can obtain relatively accurate detection results for various HSIs in a short test time, they lack physical interpretability, resulting in poor generalization. Furthermore, they still lack the use of spatial information in HSIs, and their detection results are still affected differently by anomalies and sparse backgrounds that remain in the training dataset.
3) Currently, most of the model-based detectors and all the data-driven detectors do not consider nonlinear mixing phenomena, but this is crucial in practical applications.

To solve the above problems, we propose a transformer (TR)-based AE framework (TAEF) for nonlinear hyperspectral anomaly detection, which adopts an anomaly detection strategy driven by both knowledge and data to realize the combination of model-based and data-driven anomaly detection methods, which can not only reconstruct the background more accurately to achieve high-precision anomaly detection in a short test time but also has strong physical interpretability. Specifically, different from that the encoders in most of the existing AE-based anomaly detectors consist of several linear layers, which map and reconstruct each pixel separately, and thus disregard the crucial spatial information in HSIs, we adopt TR [27] as the encoder of AE to comprehensively utilize the spatial correlation between pixels owing to the unique

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long-range self-attention (SA) mechanism of TR. The TR is the first time to serve as the encoder of AE in the hyperspectral anomaly detection task as far as we know. What is more, we embed the extended multilinear mixing model (EMLM) [28] into the decoder of AE so that the high-order nonlinear mixing phenomenon in HSIs can be characterized precisely. It should be noted that the proposed TAEF is flexible enough to characterize other degrees of mixing phenomenon by replacing the EMLM to the other mixing models. To support the TR in the characterization of the transitive global spatial information to a certain extent during the repetitive training, a novel patch generation method is proposed by the overlapped windows (OWin) in this article. Besides, to further enhance the final background reconstruction accuracy, we reduce the potential anomalies in the training set by adopting the local-clustering method [29], in which the HSIs are initially divided into several superpixel blocks by a superpixel segmentation method, and then, a clustering method is utilized and performed inside each superpixel block. By replacing the potential anomalies with sparse backgrounds, the proportion of the potential anomalies in the training set is decreased and the contamination of anomalies during the training is mitigated. In the meantime, the proportion of the sparse backgrounds in the training set is increased oppositely and the reconstruction accuracy of sparse backgrounds is enhanced. Once the training process of the proposed network is finished, the reconstruction errors of the entire hyperspectral imagery can be used to calculate the final anomaly detection result.

The main contributions of this article can be briefly summarized as follows.

1) A novel TAEF is proposed to effectively detect the anomalies in HSIs. Specifically, the TR serves as the encoder of the AE, and the EMLM is embedded into the decoder of the AE. By using the knowledge and data-driven anomaly detection strategy to realize the combination of model-based and data-driven anomaly detection methods, the proposed TAEF effectively achieves high-precision nonlinear hyperspectral anomaly detection with strong physical interpretability. It should be a pioneer work for NN-based detectors to address nonlinear mixing effects.
2) In order to support the TR to characterize the transitive global spatial information during repetitive training, an overlapped patch generation method is constructed for hyperspectral anomaly detection in this article. Besides, the local-clustering-based preprocessing by combining a superpixel segmentation method with a clustering method is adopted to adjust the proportion of potential anomalies and sparse backgrounds, which can further improve the background reconstruction accuracy of the proposed method and then achieve high-precision anomaly detection.

The rest of this article is arranged as follows. The related works are briefly reviewed in Section II. Section III describes the details of the proposed TAEF. In Section IV, the experimental results on eight real hyperspectral datasets are reported and analyzed. Finally, the conclusions are drawn in Section V.

II. Related Works

A. Spectral Mixing Models

1) LMM: Each pixel in HSIs can be regarded as the combination of several endmembers due to the low spatial resolution of HSIs [30]. Assuming that $\boldsymbol{M} = [\boldsymbol{m}_1, \ldots, \boldsymbol{m}_P] \in R^{B\times P}$ is the endmember matrix with P endmembers and B spectral bands, the most basic LMM [9] can be formulated for an arbitrary pixel $\boldsymbol{y} \in R^{B\times 1}$

$$\begin{aligned} \boldsymbol{y} &= \sum_{i=1}^{P} \alpha_i \boldsymbol{m}_i + \boldsymbol{n} \\ &\text{s.t. } \alpha_i > 0, \quad \sum_{i=1}^{P} \alpha_i = 1 \end{aligned} \tag{1}$$

where $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_P]^T$ includes the proportions of each endmember in $\boldsymbol{y}$, namely, the corresponding abundance vector of $\boldsymbol{y}$. Besides, $\boldsymbol{n}$ denotes the additive noise. It should be noted that each $\boldsymbol{\alpha}$ obeys two constraints, i.e., abundances' nonnegativity constraint (ANC) and abundances' sum-to-one constraint (ASC). Nevertheless, LMM is not sufficient to describe the nonlinear mixing phenomenon on account of the light scattering effects in complex scenes.

2) MLM: To characterize the second-order nonlinear mixing phenomenon in HSIs, several bilinear mixing models (BMMs), including the Fan model [31], the generalized bilinear model (GBM) [32], and the polynomial post-nonlinear model (PPNM) [33], are presented. Furthermore, to characterize the real high-order light scattering effects on several substances, the multilinear mixing model (MLM) [34] postulates that a beam of light reaches at least one substance and the probabilities of reaching the next substance and directly reaching the sensor are q and $(1-q)$, respectively. Then, the MLM can be formulated on the basis of the above assumption

$$\begin{aligned} \boldsymbol{y} &= (1-q)\sum_{i=1}^{P} \alpha_i \boldsymbol{m}_i + (1-q)q \sum_{i=1}^{P}\sum_{j=1}^{P} \alpha_i \alpha_j \boldsymbol{m}_i \odot \boldsymbol{m}_j \\ &\quad + (1-q)q^2 \sum_{i=1}^{P}\sum_{j=1}^{P}\sum_{k=1}^{P} \alpha_i \alpha_j \alpha_k \boldsymbol{m}_i \odot \boldsymbol{m}_j \odot \boldsymbol{m}_k + \cdots + \boldsymbol{n} \\ &= (1-q)\boldsymbol{y}_l + q\boldsymbol{y}_l \odot \boldsymbol{y} + \boldsymbol{n} \end{aligned} \tag{2}$$

where $\odot$ denotes the Hadamard product and $\boldsymbol{m}_i \odot \boldsymbol{m}_j = [m_{i,1}m_{j,1}, m_{i,2}m_{j,2}, \ldots, m_{i,B}m_{j,B}]^T$ represents the second-order scattering between two endmembers. $\boldsymbol{y}_l = \sum_{i=1}^{P} \alpha_i \boldsymbol{m}_i$ is the linear mixing component of $\boldsymbol{y}$, which corresponds to LMM. As previously mentioned, the scalar probability parameter q is utilized to measure the scattering degree of $\boldsymbol{y}$. Also, the MLM is flexible enough to contain all orders of scattering effects. Specifically, the higher the value of q, the higher the scattering degree, and vice versa. When q decreases to zero, the MLM completely degrades to LMM. However, the MLM is based on the assumption that the reflected probabilities of light in different bands are the same, which is idealistic and unpractical.

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3) EMLM: Unfortunately, the actual scattering mode in real-world scenarios is more intricate. The light has higher reflected probabilities in some specific spectral ranges because of the architectural features of scenes and the variances of light absorption properties of different substances [35], [36]. In other words, the fitting precisions in some bands are poor by setting a scalar reflected probability in all spectral bands. To solve this problem, the EMLM [28] with a vectorial $\boldsymbol{q}$ is presented as follows:

$$\boldsymbol{y} = (\mathbf{1}_B - \boldsymbol{q}) \odot \boldsymbol{y}_l + \boldsymbol{q} \odot \boldsymbol{y}_l \odot \boldsymbol{y} + \boldsymbol{n} \tag{3}$$

where $\mathbf{1}_B$ denotes a B-dimensional vector with all 1s and $\boldsymbol{q} = [q_1, q_2, \ldots, q_B]^T$ is the band-wise reflected probabilities of $\boldsymbol{y}$. That is to say, the reflected probabilities in different bands are different, which provides enough degrees of freedom to accurately characterize the nonlinear mixing phenomenon in HSIs. It should be noted that although the reflected probability $\boldsymbol{q}$ varies in the band dimension, the adjacent bands often have similar degrees of reflected probabilities [37].

B. AE

The AE [21] has been widely used in various applications such as unmixing [38], [39] and target detection [40], [41] because of the simple structure and outstanding performance in feature extraction. Take the simplest AE for example to explain the general processes of AE, in which the encoder and the decoder are both a fully connected layer. For an arbitrary input $\boldsymbol{y} \in R^{B\times 1}$, the hidden layer $\boldsymbol{z} \in R^{L\times 1}$ with L ($L < B$) dimensions and the reconstructed output $\hat{\boldsymbol{y}} \in R^{B\times 1}$ can be calculated as follows:

$$\boldsymbol{z} = f(\boldsymbol{W}_{\text{en}}\boldsymbol{y} + \boldsymbol{b}_{\text{en}}) \tag{4a}$$

$$\hat{\boldsymbol{y}} = f(\boldsymbol{W}_{\text{de}}\boldsymbol{z} + \boldsymbol{b}_{\text{de}}) \tag{4b}$$

where $\boldsymbol{W}_{\text{en}}$ and $\boldsymbol{b}_{\text{en}}$ and $\boldsymbol{W}_{\text{de}}$ and $\boldsymbol{b}_{\text{de}}$ denote the weights and biases of the encoder and decoder, respectivel, and $f(\cdot)$ is an inconclusive activation function. It is worth noting that several nonlinear activation functions, i.e., sigmoid and softmax, can introduce nonlinearity through mapping data but cannot characterize the inherent nonlinear interactions between endmembers in each hyperspectral pixel. AE can automatically extract the deep and abstract features through the processes of dimensionality reduction and reconstruction of all inputs and thus has a certain robustness to random noise. In different applications, the specific structures of the encoder and decoder of AE would require some targeted modifications.

As stated previously, the rationale for using AE to handle the hyperspectral anomaly detection task is the reality that the overwhelming majority of pixels in HSIs belong to the background and only a few pixels are anomalies. From the aspect of feature extraction, AE progressively emphasizes the background features during the training process, and thus, the finally reconstructed background pixels are as similar as possible to the inputs. Simultaneously, the anomaly features are gradually ignored, which results in significant reconstruction errors of anomalies. Therefore, the reconstruction error of a well-trained AE can be used to calculate the final anomaly detection result.

C. TR

Owing to the unique long-range SA mechanism, TR [27] has been widely used in various vision tasks, such as image segmentation [42], [43] and classification [44], [45]. The traditional TR module consists of three components, i.e., a multiple layer perception (MLP), a multihead SA (MSA), and a layer normalization (LN). The MSA block is formed by concatenating several SA blocks, which is the core of the SA mechanism in TR. For an arbitrary input $\boldsymbol{Y} \in R^{B\times N}$ with N pixels, the corresponding output after passing an SA block is formulated as follows:

$$\boldsymbol{Q} = \boldsymbol{W}_q\boldsymbol{Y} \in R^{L\times N} \tag{5a}$$

$$\boldsymbol{K} = \boldsymbol{W}_k\boldsymbol{Y} \in R^{L\times N} \tag{5b}$$

$$\boldsymbol{V} = \boldsymbol{W}_v\boldsymbol{Y} \in R^{L\times N} \tag{5c}$$

$$\boldsymbol{SA} = \text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{B}}\right)\boldsymbol{V} \in R^{L\times N} \tag{5d}$$

where $\boldsymbol{W}_q \in R^{L\times B}$, $\boldsymbol{W}_k \in R^{L\times B}$, and $\boldsymbol{W}_v \in R^{L\times B}$ are three pre-set transformation matrices. It is worth noting that the transformation matrices in each SA block are independent. Then, the results of H heads can be concatenated by $\boldsymbol{W} \in R^{B\times HL}$ as follows:

$$\text{MSA}(\boldsymbol{Y}) = \boldsymbol{W}[\boldsymbol{SA}_1; \ldots; \boldsymbol{SA}_H] \in R^{B\times N}. \tag{6}$$

It is evident from (5) that the TR considers the relationships between each input and all other inputs. Especially for HSIs, the TR is good at fully excavating the spatial information, which is important, but it is lacking for most NN-based anomaly detectors.

III. Proposed Method

As stated previously, AE progressively emphasizes the background features during the training process. In other words, the background can be reconstructed well by AE, whereas anomalies cannot. The proposed TAEF is directed at strengthening the difference of the reconstruction results, namely, improving the reconstruction results of background and suppressing the reconstruction results of anomalies. Specifically, the proposed TAEF is mainly composed of the following parts: a TAEF with TR encoder and EMLM embedded decoder, a patch generation method that supports the TR in the characterization of local-to-global spatial information, and a local-clustering method that reduces the potential anomalies while adding the sparse backgrounds. The overall graphical description of the proposed TAEF is illustrated in Fig. 1.

A. TR-Based AE

1) TR Encoder: Let $\boldsymbol{Y} \in R^{B\times N}$ denote the original 2-D dataset, namely, the testing dataset. Then, the training dataset $\boldsymbol{Y}_1 \in R^{B\times N}$ can be obtained by decreasing the potential anomalies and meanwhile increasing the sparse backgrounds by the local-clustering method. To extract only the features of background, the training goal of the proposed TAEF is to learn the features of training dataset $\boldsymbol{Y}_1$ and reconstruct $\boldsymbol{Y}_1$ as similar

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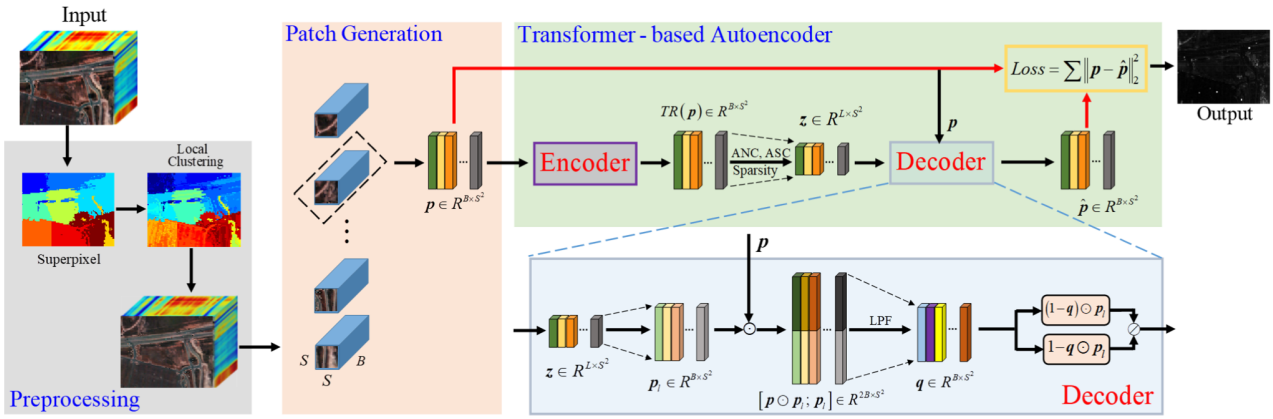

Fig. 1. Overview of the proposed method for nonlinear hyperspectral anomaly detection.

as possible. For the sake of computation efficiency, the training dataset $\boldsymbol{Y}_1$ is divided into many patches by the OWin before feeding it into the TR encoder. The specific procedures of the overlapped patch generation method and the local-clustering method will be given and described in subsequent subsections.

As summarized above, the spatial information between pixels is not considered in most of the existing AE-based anomaly detectors. However, the spatially adjacent pixels tend to be similar on account of the low spatial resolution of HSIs [46]. In other words, each pixel can be represented by its neighboring pixels, which is also the cornerstone of CRDs. To this issue, the TR is utilized to play the role of encoder owing to its unique SA mechanism, which can fully excavate the spatial similarity between pixels and realizes the data-driven strategy. Then, the TR encoder can more effectively extract the features of background component, which finally benefits the reconstruction results of the whole network.

Without loss of generality, take an arbitrary patch $\boldsymbol{p} \in R^{B\times S^2}$ (here, the patch size is $S \times S$) for example to give the specific process of the TR encoder.

1) A random position code $\boldsymbol{pc}$ is added to describe the position information in the SA mechanism:

$$\boldsymbol{p}' = \boldsymbol{p} + \boldsymbol{pc} \in R^{B\times S^2}. \tag{7}$$

2) Obtain the output MSA($\boldsymbol{p}'$) of TR as (6).
3) Compute the hidden layer of AE by a linear layer

$$\boldsymbol{z} = f\left(\boldsymbol{W}_z \mathrm{MSA}\left(\boldsymbol{p}'\right)\right) \in R^{L\times S^2} \tag{8}$$

where $\boldsymbol{W}_z \in R^{L\times B}$ is the weight matrix of the linear layer. The activation function $f(\cdot)$ is the softmax function, which makes the output $\boldsymbol{z}$ compliant with the ANC and ASC constraints to obtain accurate abundances and reconstructed pixels. Moreover, the Napierian logarithm inside the softmax function also imposes sparsity to $\boldsymbol{z}$, which is consistent with the fact that each pixel consists of only a few endmembers.

2) EMLM Embedded Decoder: All existing AE-based anomaly detectors can only characterize linear mixing phenomenon because, in these anomaly detectors, all decoders consist of linear layers. To characterize the ubiquitous nonlinear mixing phenomenon, EMLM is embedded into the decoder in this article. Note that (3) can be converted to the following form by changing the pixel $\boldsymbol{y}$ to the corresponding patch $\boldsymbol{p}$:

$$\hat{\boldsymbol{p}} = \frac{(\boldsymbol{1}_{B\times S^2} - \boldsymbol{q}) \odot \boldsymbol{p}_l}{\boldsymbol{1}_{B\times S^2} - \boldsymbol{q} \odot \boldsymbol{p}_l}. \tag{9}$$

It can be seen that the integrated nonlinear mixing result can be calculated by its linear component $\boldsymbol{p}_l$ and the reflected probability matrix $\boldsymbol{q}$. Therefore, we still adopt the fully connected layer to obtain the linear component $\boldsymbol{p}_l$ such as other AE-based anomaly detectors

$$\boldsymbol{p}_l = \boldsymbol{W}_l \boldsymbol{z} \in R^{B\times S^2}. \tag{10}$$

Then, the stacked features $[\boldsymbol{p} \odot \boldsymbol{p}_l, \boldsymbol{p}_l]$, which include the original spectral curves and their linear component, are utilized to obtain the reflected probability matrix $\boldsymbol{q}$ through other fully connected layers. It is worth noting that the spectral curve of each pixel is almost smooth and the values of adjacent bands are close because of the continuity of the light reflection characteristics. Likewise, the reflected probability matrix $\boldsymbol{q} = [q_1, q_2, \ldots, q_B]^T$ is also smooth in band dimension [36]. Based on this, for simplicity, a low-pass filter (LPF) can be added behind the fully connected layer, e.g., with a bandwidth set to 3, as follows:

$$\begin{aligned} \hat{q}_1 &= \frac{1}{3}(q_1 + 2q_2) \\ \hat{q}_i &= \frac{1}{3}\sum_{j=i-1}^{j=i+1} q_j, \quad i = 2, \ldots, B-1 \\ \hat{q}_B &= \frac{1}{3}(q_B + 2q_{B-1}). \end{aligned} \tag{11}$$

Once the linear component $\boldsymbol{p}_l$ and the reflected probability matrix $\hat{\boldsymbol{q}}$ are obtained, the final nonlinear mixing result can be calculated by (9). In this manner, the pixels are reconstructed under the constraint of a certain mixing model, which introduces strong physical interpretability and enables the model-based strategy to be implemented. Due to the embedding of the physical knowledge (spectral mixture model) into the decoder, the solution space can be limited by the embedded physical knowledge (specifically EMLM), which can make the proposed method to have better generalization, so that its background reconstruction performance is better.

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TABLE I
DETAILED CONFIGURATIONS OF THE PROPOSED NETWORK

Network		Parameter	Value
TR Encoder	Overlapped Patches	Size S	10
	TR	Head	6
		Hidden Channel of MLP	10
	Fully connected layer 1	Input Channel	B
		Output Channel	10
EMLM embedded decoder	Fully connected layer 2	Input Channel	10
		Output Channel	B
	Fully connected layer 3	Input Channel	$2B$
		Output Channel	B

It should be noted that other mixing models can also be embedded into the AE decoder similarly, and thus, the proposed AE framework is flexible enough to characterize various mixing phenomena by simply adjusting the decoder of AE. Besides, while the TR encoder and the EMLM embedded decoder play a role in characterizing the local-to-global spatial information in HSIs and introducing physical interpretability, respectively. They are tightly connected by the hidden layer and trained together by a loss function, i.e., to achieve joint utilization of abundances.

3) Loss Function: Equation (12) gives the training loss function, and the specific configurations of our network are listed in Table I. After repeating the training process on the training dataset $\boldsymbol{Y}_1$ in many epochs, the network can fully learn the background features and accurately reconstruct the background

$$\text{Loss} = \sum_{i=1}^{N} \left\| \boldsymbol{y}_i - \hat{\boldsymbol{y}}_i \right\|_2^2. \tag{12}$$

Finally, the testing dataset $\boldsymbol{Y}$ is also fed into the well-trained network. Owing to the outstanding background reconstruction capability of the proposed TAEF and the proportion adjustment of the potential anomalies and the sparse backgrounds by the local-clustering method, the background in testing dataset $\boldsymbol{Y}$ can be reconstructed effectively, whereas the reconstruction of anomalies is terrible. Therefore, the reconstruction errors can be regarded as the anomalous levels

$$\left\| \Delta \boldsymbol{y}_i \right\|_2^2 = \left\| \boldsymbol{y}_i - \hat{\boldsymbol{y}}_i \right\|_2^2 = \sum_{b=1}^{B} \left(\boldsymbol{y}_{ib} - \hat{\boldsymbol{y}}_{ib} \right)^2. \tag{13}$$

B. Patch Generation Method

As described previously, the long-range SA mechanism can characterize the relationships between pixels. However, there are usually a large number of pixels in HSIs, and it is time-consuming to feed all pixels into a TR module at the same time. Therefore, a reasonable solution is to divide the HSIs into many patches, in which the number of pixels in each patch is controllable and the time cost of the TR processing becomes acceptable.

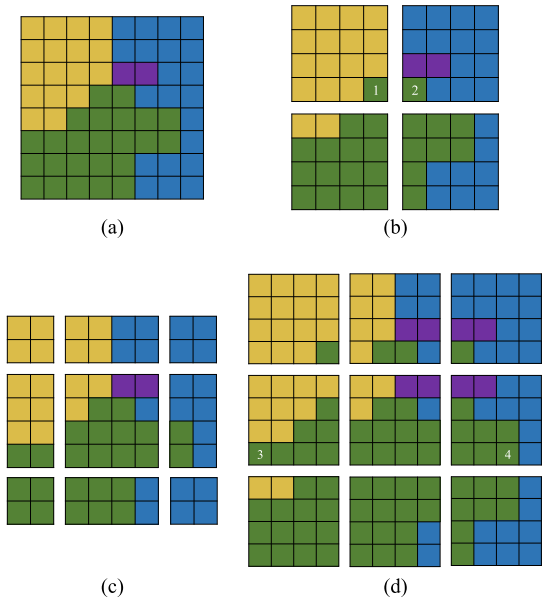

Fig. 2. Illustrations of different patch generation methods, in which the green, yellow, and blue pixels denote three categories of backgrounds, and the purple pixels are anomalies. (a) Sketch map with 8 × 8 pixels for simplified hyperspectral imagery with four ground objects. (b) Patches produced by traditional patch generation method (i.e., equivalent to the first stage of SWin). (c) Patches produced by the second stage of SWin. (d) Patches produced by the overlapped window (OWin).

Let the sketch map with 8 × 8 pixels denote simplified hyperspectral imagery in Fig. 2. The patches produced by the traditional patch generation method do not overlap each other, which results in losing the relationship between two adjacent pixels, such as the 1st and 2nd pixels in Fig. 2(b). To address this problem, the high-profile SWin-TR [47] proposes a two-stage patch generation method that connects the traditional patch generation method shown in Fig. 2(b) and the shift window (SWin) shown in Fig. 2(c) so that the relationship between any two adjacent pixels can be considered.

However, in comparison to SWin-TR used in the natural images, there is another problem in the case of HSIs, that is, the small patch in Fig. 2(c) only contains a few pixels, which affects the characterization of spatial information. Therefore, we introduce the OWin shown in Fig. 2(d) to support the TR to characterize the global spatial information. It should be noted that the OWin used is also known as a sliding window with a certain interval that is set to half the patch size so that the windows overlap each other. In this case, even the relationship between two distant pixels, such as the 3rd and 4th pixels in Fig. 2(d), can be effectively characterized over many training iterations. Therefore, the global spatial information is also indirectly considered in this article. Similar to SWin-TR, the combination of the utilized patch generation method and the TR encoder is named OWin-TR.

C. Local-Clustering-Based Preprocessing

AE has an ability to automatically learn the background features, which is the cornerstone of using AE in the HSI anomaly detection task. However, there are two obvious problems in the existing AE-based anomaly detectors: 1) the anomalies with a significantly small proportion still affect

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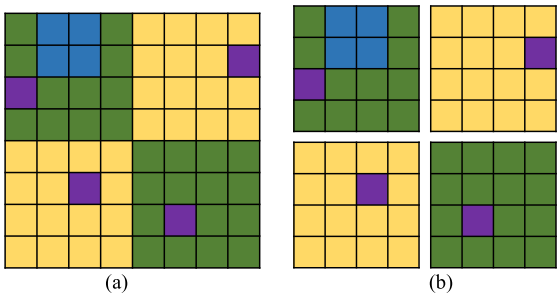
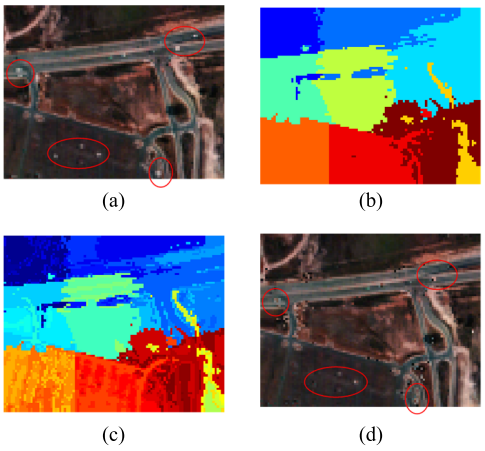


(a) (b)

Fig. 3. Difference of the anomalies (purple pixels) and sparse backgrounds (blue pixels). The green pixels and the yellow pixels denote two categories of backgrounds. (a) Sketch map with 8 × 8 pixels for simplified hyperspectral imagery with four ground objects. (b) Superpixel segmentation map produced by the superpixel method.



(a) (b) (c) (d)

Fig. 4. Visual illustration of the local-clustering method on the HYDICE dataset. (a) Pseudocolor image of the original dataset. (b) Preliminary superpixel segmentation map obtained by a superpixel method. (c) Final clustering map obtained by a clustering method on each superpixel block. (d) Pseudocolor image of the preprocessed dataset.

the training process of AE and 2) the presence of sparse backgrounds in HSI, even if they account for a small proportion, will lead to poor background reconstruction using AE, resulting in a serious false alarm rate in the anomaly detection map. Several anomaly detectors have solved the first problem by predetection to a certain extent [25], [48], whereas the second problem has never been considered before.

To solve the above two problems in the meantime, we adopt the local-clustering method [29] for preprocessing, which decreases the potential anomalies and meanwhile increases the sparse backgrounds in the training dataset. Take the 8 × 8-pixel sketch in Fig. 3(a) as an example, and the proportions of anomalies (purple pixels) and the sparse backgrounds (blue pixels) in the whole image are all 6.25%, which cannot be distinguished by a clustering method. After applying a superpixel segmentation method, the whole sketch is first divided into four superpixel blocks as shown in Fig. 3(b), where the anomalous proportion of each superpixel block remains at 6.25%, but the proportion of the sparse backgrounds in the upper left superpixel block increases to 25%. After that, the potential anomalies and sparse backgrounds can be fully distinguished by applying a clustering method in each superpixel block.

Moreover, we demonstrate the visual effect of the proposed preprocessing using the Hyperspectral Digital Imagery Collection Experiment (HYDICE) dataset that will be detailed in Section IV. As can be clearly seen from Fig. 4(a), the white ground objects in the right bottom with a relatively small proportion is the sparse background. The preliminary superpixel segmentation map and the final clustering map are shown in Fig. 4(b) and (c), respectively. Although the shape of the generated superpixel blocks is irregular due to the characteristics of the superpixel method used, the sparse background pixels are still grouped together and can be distinguished from the potential anomalies. Subsequently, the location of these potential anomalies is replaced by the sparse background pixels, as shown in Fig. 4(d). In this manner, the proportion of anomalies is decreased and the proportion of sparse backgrounds is increased in the training dataset $\boldsymbol{Y}_1$. Consequently, the above two problems can be solved well at the same time, which significantly benefits the background reconstruction in AE.

The entire process of the proposed TAEF can be summarized in Algorithm 1.

Algorithm 1 Entire Process of TAEF

Input: Dataset $\boldsymbol{Y} \in R^{B\times N}$.

Initialize: Obtain the training dataset $\boldsymbol{Y}_1 \in R^{B\times N}$ by the local clustering-based preprocessing.

Patch Generation: Generate overlapped patches by the OWin patch generation method.

For each epoch, **do**

1) Map each patch into the hidden layer with (8).
2) Reconstruct each patch with (9).
3) Train the network by minimizing *Loss* in (12).

End for

Output: obtain detection map with (13).

IV. Experimental Results and Analysis

The performance of the proposed TAEF is evaluated adequately in this section. Specifically, the ablation experiments are first executed to verify the effectiveness of each component in the proposed TAEF. Then, the bandwidth of the LPF is tested and analyzed. Third, the performances of the proposed TAEF and other state-of-the-art (SOTA) anomaly detectors are compared and analyzed on eight real hyperspectral datasets. Finally, the robustness of the proposed TAEF is measured by noise experiments.

All the experiments are implemented by Pycharm 2021.3.2 and MATLAB R2018b on a PC with 32-GB RAM and an Intel Core i9-11900K CPU. The epochs, learning rate, and batch size are set to 20, 0.005, and 100, respectively. It is worth noting that simple linear iterative clustering (SLIC) [49], which comprehensively considers the spectral–spatial distance between pixels, is adopted to obtain an outstanding superpixel segmentation result in preprocessing. Besides, the density peak-based clustering method (DPBC) [50], which avoids the presetting of the number of clusters, is adopted as the clustering method in preprocessing.

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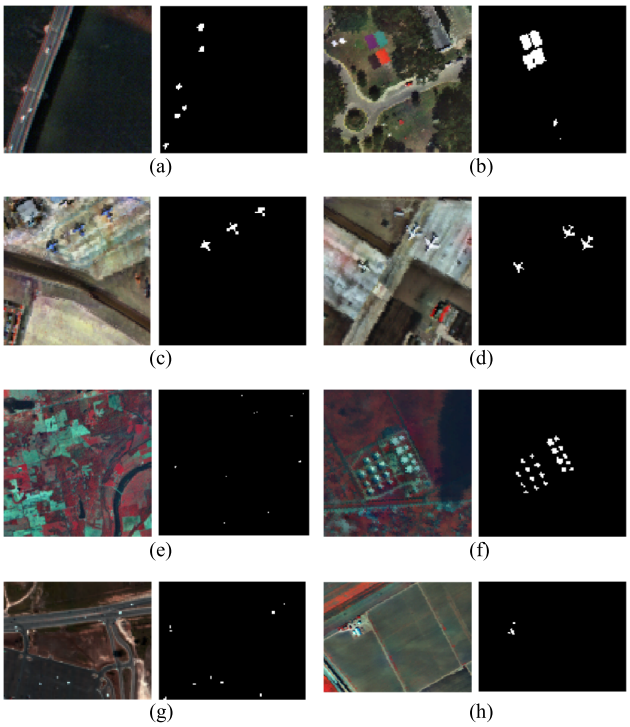

Fig. 5. Pseudocolor images (left) and the corresponding ground-truth maps (right) of eight real hyperspectral datasets. (a) Pavia. (b) MUUFL. (c) San Diego I. (d) San Diego II. (e) Hyperion. (f) Texas Coast. (g) HYDICE. (h) Salinas.

A. Datasets Descriptions

In the experiments of comparison with other SOTA competitors, eight widely used real hyperspectral datasets are employed. The pseudocolor images and the corresponding ground-truth maps of these datasets are displayed in Fig. 5. It is evident that the anomalies in these datasets are varied in terms of type and size, which can be used to evaluate and validate the anomaly detection performance of the proposed TAEF in a variety of different situations.

1) *Pavia:* This real dataset was collected in Pavia, Italy, by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor [51]. The original Pavia dataset contains 1096 × 1096 pixels with 102 bands and its spatial resolution is 1.3 m/pixel. In this article, a subimage with 100 × 100 pixels is cropped in the following experiments, and the vehicles on the bridge are considered as anomalies. It should be noted that this hyperspectral imagery contains significantly the high-order scattering effects because of the prominent altitude difference between vehicles, bridge, and river.
2) *MUUFL:* This real dataset was captured at the University of Southern Mississippi Gulf Park Campus, USA, by the Compact Airborne Spectrographic Imager (CASI) sensor [52]. The original size of the MUUFL dataset is 325 × 220 with 64 spectral bands, and a 100 × 100 subimage is adopted in the following experiments. The anomalies to be detected are four artificially placed cloth panels. Similarly, the high-order scattering effects are also widely existed in this dataset due to the prominent altitude difference between buildings, trees, and ground.
3) *San Diego:* This real dataset was collected in San Diego, USA, by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [53]. The original hyperspectral imagery contains 400 × 400 pixels with 224 bands, and only 186 bands are reserved in this article after eliminating several low-grade bands. Two subimages with 100 × 100 pixels are cropped in our experiments, and the airplanes are considered as anomalies. Due to the low spatial resolution of HSIs, a number of subpixel anomalies exist in the edges of airplanes and ground, which results in considerable nonlinear scattering effects.
4) *Hyperion:* This real dataset was collected in an agricultural region, USA, from the EO-1 satellite image website [54]. The original Hyperion dataset contains 150 × 150 pixels with 242 bands and its spatial resolution is 10 m/pixel. In this article, 155 bands are reserved after removing some low signal-to-noise ratio (SNR) bands. The anomalies to be detected in this dataset are the storage silos.
5) *Texas Coast:* This real dataset was captured at the Texas Coast, USA, by the AVIRIS sensor [51]. This hyperspectral imagery contains 100 × 100 pixels with 207 bands and its spatial resolution is 17.2 m/pixel. The buildings are the anomalies to be detected in this dataset.
6) *HYDICE:* This real dataset was collected at a residential area by the HYDICE sensor. The original size of the HYDICE dataset is 400 × 400 with 174 spectral bands, and an 80 × 100 subimage is cropped in the following experiments. The vehicles and roofs are considered as the anomalies in this dataset.
7) *Salinas:* This real dataset covers the Salinas Valley, USA, by the AVIRIS sensor [55]. The original dataset contains 512 × 217 pixels with 224 bands and its spatial resolution is 3.7 m/pixel. After removing the irrelevant regions and the low-grade bands, a subimage with 200 × 150 pixels and 204 bands is adopted in the experiments. The anomalies to be detected in this dataset are the artificial structures.

B. Evaluation Metrics

In this article, the detection accuracy of all anomaly detectors is evaluated comprehensively by the 3-D receiver operating characteristic (3D ROC) curve of (P_D, P_F, τ) [56], from which three 2-D ROCs (i.e., $\text{ROC}_{(D,F)}$, $\text{ROC}_{(D,\tau)}$, and $\text{ROC}_{(F,\tau)}$) and the corresponding AUCs (i.e., $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$) can be obtained. Besides, five other AUCs can also be calculated by the above three AUCs as follows:

$$\text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)} \tag{14a}$$

$$\text{AUC}_{\text{TD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} \tag{14b}$$

$$\text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}} \tag{14c}$$

$$\text{AUC}_{\text{TD-BS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \tag{14d}$$

$$\text{AUC}_{\text{OD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \tag{14e}$$

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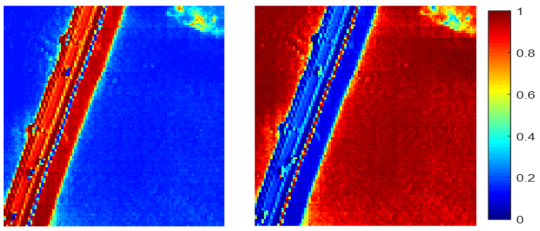

(a) (b)

Fig. 6. Features (the hidden layer output of the AE) extracted by the TR encoder from the Pavia dataset, which correspond to the abundance maps of (a) bridge and (b) river.

TABLE II
DETAILED SETTINGS OF DIFFERENT CASES IN ABLATION EXPERIMENTS

Ablation Cases		Vectorized Reflected Probability $\boldsymbol{q}$	Loss-pass Filter	Patch Generation Method	Nonlinear Mixing Model
Ablation on Different Components	C1	×	√	√	√
	C2	√	×	√	√
	C3	√	√	×	√
	C4	√	√	√	×
	C5	√	√	√	√

where $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{\mathrm{TD\text{-}BS}}$, and $\mathrm{AUC}_{\mathrm{OD}}$ are utilized to assess the overall accuracy of detectors. $\mathrm{AUC}_{(F,\tau)}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$ are utilized to assess the background suppressibility of detectors. $\mathrm{AUC}_{(D,\tau)}$ and $\mathrm{AUC}_{\mathrm{TD}}$ are utilized to assess the anomaly detectability of detectors. It is worth noting that a smaller $\mathrm{AUC}_{(F,\tau)}$ indicates a better detector, and other AUC values do the opposite.

Due to the difficulty of measuring the computational complexity of all employed anomaly detectors, the time consumption is utilized to compare the computational speed. Note that there are two stages, i.e., training and test, in NN-based detectors, and these two stages can be executed separately. Moreover, the dataset can be tested repeatedly on a well-trained network. Therefore, it is necessary to list the test time of NN-based detectors separately.

C. Middle Results of the Proposed TAEF

Actually, the AE framework implements the unmixing process of each pixel in the HSIs, and thus, the features extracted by the TR encoder (the hidden layer output of the AE) correspond to the abundances of HSIs. Take the first dataset (Pavia) for example, and the features (the hidden layer output) extracted by the TR encoder are shown in Fig. 6, which correspond to the abundance maps of bridge and river.

D. Ablation Experiment

Here, several ablation experiments are conducted to verify the effects and benefits of the involved components in the proposed TAEF. Specifically, five cases in the ablation experiments are listed in Table II.

The anomaly detection maps of all the ablation cases and the corresponding 3D-ROC curves are displayed in Figs. 7 and 8, respectively. Besides, the three 2D-AUC values and the five derived AUC values are listed in Table III. It can be seen from Fig. 7 that the false alarms in the detection maps of C1, C2, C3, and C4 are obviously more than that of C5, which can be further concluded from Table III that the background suppressibility AUCs (i.e., $\mathrm{AUC}_{(F,\tau)}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$) of C5 are better than those of C1, C2, C3, and C4. Compared with the EMLM model in C5, C1, C2, and C4 assume the reflected probability $\boldsymbol{q}$ as a scalar, a band-independent vector, and a vector with all 0s, respectively. It is reasonable to suppose that the background reconstruction accuracy of network is affected based on the above three imprecise assumptions. In comparison to the OWin in C5, the patches in C3 are independent of each other and do not overlap, which limits the utilization of spatial information between pixels by TR and affects the background reconstruction accuracy of network. Comparatively speaking, the C5 has a better background reconstruction accuracy owing to the accurate characterization of nonlinear mixing effects by EMLM and the sufficient utilization of the local-to-global spatial information by OWin-TR. On the contrary, the anomaly detectability AUCs (i.e., $\mathrm{AUC}_{(D,\tau)}$ and $\mathrm{AUC}_{\mathrm{TD}}$) of C5 are less than ideal between five cases due to a large number of false alarms in C1, C2, C3, and C4. However, C5 has the highest $\mathrm{AUC}_{(D,F)}$ on the two datasets and the highest $\mathrm{AUC}_{\mathrm{TD\text{-}BS}}$ and $\mathrm{AUC}_{\mathrm{OD}}$ on the MUUFL dataset, which demonstrates the benefits of the involved components in the proposed TAEF.

In the following various experiments, only the $\mathrm{ROC}_{(D,F)}$ curves and the $\mathrm{AUC}_{(D,F)}$ values are utilized to evaluate the overall accuracy of anomaly detectors for the sake of conciseness.

E. Parameter Setting

U denotes the bandwidth of LPF, which corresponds to the smooth level of the reflected probability $\boldsymbol{q}$. To make the bandwidth of LPF symmetrical around the center band, the value of U is set as odd numbers, i.e., [3, 5, 7, 9, and 11].

The $\mathrm{AUC}_{(D,F)}$ values of different U are displayed in Fig. 9. It can be observed that the detection accuracy of the proposed TAEF is insensitive to the bandwidth of the LPF to some extent. Particularly in the case of the Pavia dataset, the $\mathrm{AUC}_{(D,F)}$ curve is almost a straight line. Therefore, the bandwidth of LPF U is fixed at 7 in the following experiments after fully considering the experimental results of Fig. 9 as well as the computational complexity.

F. Comparison With Other SOTA Competitors

We performed experiments and compared our TAEF with an LMM-based anomaly detector (GTVLRR [15]), two kernel-based nonlinear anomaly detectors (KIFD [18] and KNUD [20]), and four AE-based linear anomaly detectors (LREN [22], GAED [26], RGAE [23], and AutoAD [24]) to comprehensively evaluate and validate the performance of the proposed TAEF. Considering the spatial–spectral properties of HSIs, the TV and graph regularizations are incorporated in GTVLRR. KIFD detects anomalies in the high-dimensional space by applying isolated forest. KNUD executes the anomaly detection according to the endmember-kernel theory with a dual window to characterize the nonlinear mixing effects between endmembers. LREN and GAED incorporate the LRR and the guided map into AE to implement anomaly

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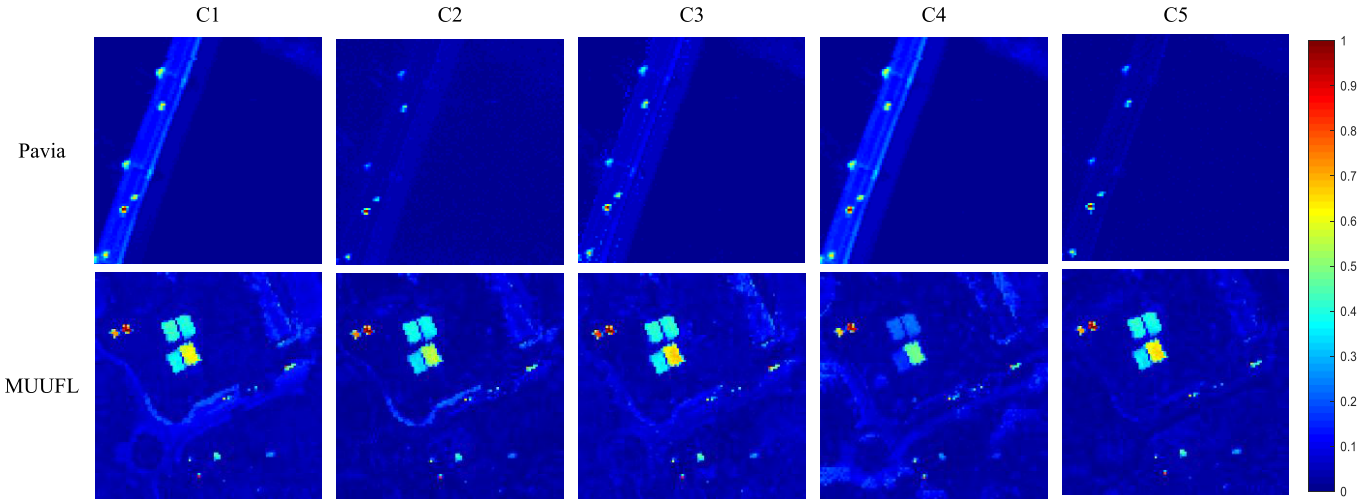

Fig. 7. Anomaly detection maps obtained by the proposed TAEF on different ablation cases.

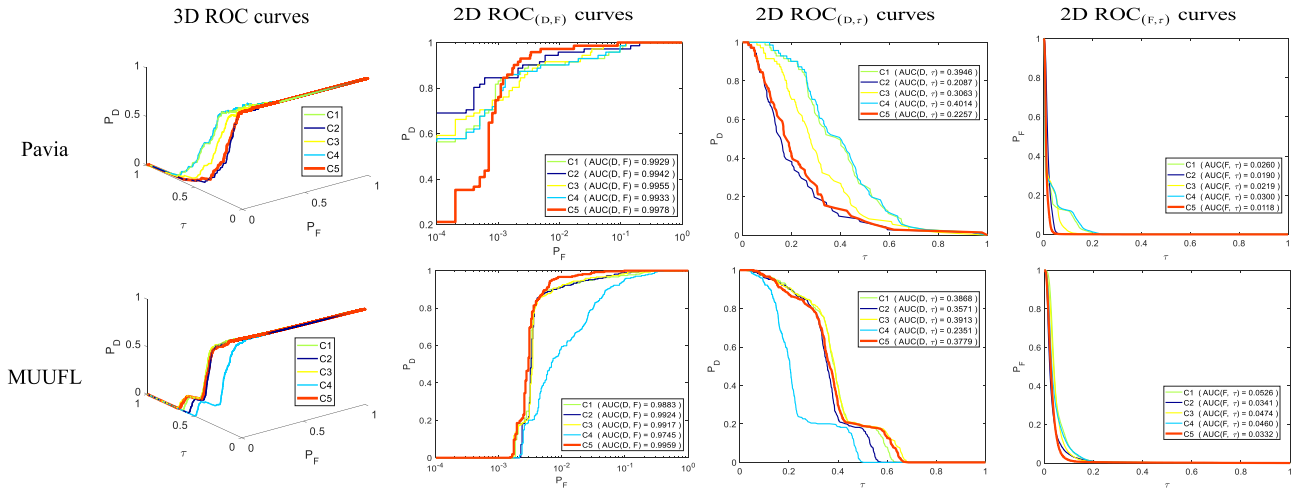

Fig. 8. 3-D ROC curves and three 2-D ROC curves on the Pavia and MUUFL datasets.

TABLE III

3-D AUC VALUES OF THE PROPOSED TAEF ON DIFFERENT ABLATION CASES

Cases		$AUC_{(D,F)}\uparrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{TD}\uparrow$	$AUC_{BS}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{TDBS}\uparrow$	$AUC_{OD}\uparrow$
Pavia	C1	0.9929	0.3946	0.0260	1.3875	0.9670	15.2045	0.3686	1.3615
	C2	0.9942	0.2087	0.0190	1.2028	0.9752	10.9614	0.1896	1.1838
	C3	0.9955	0.3063	0.0219	1.3018	0.9736	13.9982	0.2845	1.2799
	C4	0.9933	**0.4014**	0.0300	**1.3947**	0.9632	13.3620	**0.3714**	**1.3647**
	C5	**0.9978**	0.2257	**0.0118**	1.2235	**0.9860**	**19.1194**	0.2139	1.2117
MUUFL	C1	0.9883	0.3868	0.0526	1.3751	0.9357	7.3517	0.3342	1.3225
	C2	0.9924	0.3571	0.0341	1.3496	0.9584	10.4806	0.3230	1.3155
	C3	0.9917	**0.3913**	0.0474	**1.3830**	0.9443	8.2540	0.3439	1.3356
	C4	0.9745	0.2351	0.0460	1.2096	0.9285	5.1115	0.1891	1.1636
	C5	**0.9959**	0.3779	**0.0332**	1.3738	**0.9626**	**11.3783**	**0.3447**	**1.3406**

detection under LMM, respectively. RGAE imposes the graph regularization on the latent layer of AE to consider the spectral relationship between pixels, and GAED builds the AE by the fully convolutional layers to consider the spatial relationship between pixels.

The involved parameters in GTVLRR are $\lambda = 0.5$, $\beta = 0.2$, and $\gamma = 0.05$ as recommended in [15]. After extensive searching KNUD, the optimal dual window sizes are set to (3, 15), (3, 5), (3, 7), (3, 7), (3, 11), (3, 7), (3, 13), and (7, 9) for these eight real datasets. Besides, the proportion of the selected pixels in window and the quantity of anomaly atoms are set to 95% and 15 as recommended in KNUD [20], respectively. According to [18], the subsampling size and the number of trees in KIFD are set to 3% and 1000, respectively. The codes of LREN [22], GAED [26], RGAE [23], and AutoAD [24] are available from the authors and all the parameter settings follow their corresponding references for a fair comparison.

The anomaly detection maps of all datasets are shown in Fig. 10. Take the first dataset (Pavia) for example, and the high-order nonlinear mixing effects obviously exist in this dataset due to significant differences in height between

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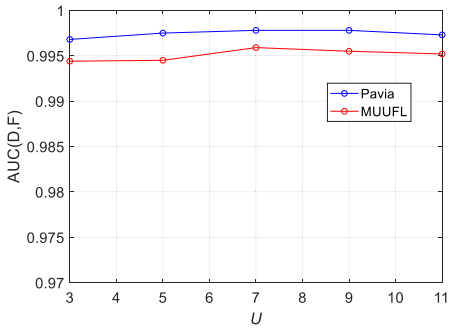

Fig. 9. $AUC_{(D,F)}$ values of the different bandwidth of LPF U on the Pavia and MUUFL datasets.

river, bridge, and vehicles. It is evident from Fig. 10 that the LMM-based detectors, i.e., GTVLRR, LREN, GAED, RGAE, and AutoAD, cannot detect all anomalies in this dataset, especially at the edges of the vehicles, which are the main locations of the nonlinear mixing effects. Inversely, the kernel-based detectors, i.e., KIFD and KNUD, can detect all anomalies. However, there are a large number of false alarms in the detection maps of KIFD due to the unreasonable nonlinear strategy, which simply maps each pixel into a high-dimensional space. The phenomenon of false alarm is more obviously reduced in KNUD because of the utilization of the endmember-kernel theory. On the contrary, owing to the strong physical interpretability and accompanying excellent background reconstruction capability, the proposed TAEF can not only effectively suppress the background but also successfully detect almost all anomalies. Another example is the Texas Coast dataset; similarly, the GTVLRR, LREN, GAED, RGAE, and AutoAD still cannot detect the locations of all anomalies, especially the group of buildings on the right. Also, there are different levels of false alarms in KIFD and KNUD. Contrarily, all anomalies can be detected by the proposed TAEF with a slight false alarm owing to the accurate characterization of nonlinear mixing effects by EMLM and the sufficient utilization of the local-to-global spatial information by OWin-TR. It should be noted that the pixel proportions of two ground objects in the upper left corner in MUUFL dataset are lower than those of the artificial anomalies to be detected, resulting in serious false alarms in the detection maps of all detectors. Leave aside this common mistake, the proposed TAEF achieves the best anomaly detection results under good background suppressions.

The $ROC_{(D,F)}$ curves and the $AUC_{(D,F)}$ values of the eight datasets are shown in Fig. 11 and Table IV, respectively. The largest value and the second largest value in each line of Table IV are bolded and underlined, respectively. It is worth noting that the abscissas in Fig. 11 are all logarithmic abscissas for facilitating visual observation, which results in the situation that the front half part of the abscissa only includes 1% scope of P_F. Hence, it can be observed that the proposed TAEF is superior to the competitors on the whole in Fig. 11(a), (b), (e), (f), and (h). Although it is hard to intuitively indicate the optimal method in the $ROC_{(D,F)}$ curves of Fig. 11(c), (d), and (g), P_D reaches 1 in a relatively small P_F in the proposed TAEF. Besides, most of the $AUC_{(D,F)}$ values of the proposed TAEF are obviously higher than those of the competitors, and only the $AUC_{(D,F)}$ value of the proposed TAEF in the HYDICE dataset is slightly smaller than that of KIFD due to the overcomplicated ground environment in the HYDICE dataset, which also verifies the superior of the proposed TAEF quantitatively.

Considering the demand of the practical application, Table IV also provides the running times of all anomaly detectors on the Pavia dataset. It can be seen that the GTVLRR is time-consuming due to the iterative optimization in plenty of iterations. Besides, the two kernel-based detectors, i.e., KIFD and KNUD, are all time-consuming because of the nonlinear mapping of the datasets. Comparatively speaking, although the training times of the AE-based detectors, i.e., LREN, GAED, RGAE, and AutoAD, are different, their test times are relatively short in the fine-trained networks. As for the proposed TAEF, the spatial relations between pixels are fully considered by the long-range SA mechanism of TR, which results in a slightly long test time but still in an acceptable range. Besides, the amount of parameters and the mega floating-point operations (MFLOPs) of the proposed TAEF on the Pavia dataset are 0.15M and 15.16, respectively.

G. Robustness to Different Levels of Noise

Here, the experiments about noise are conducted on the MUUFL dataset to measure the robustness of the proposed TAEF. Specifically, the noisy datasets are generated by corrupting different levels of Gaussian noise (30, 25, and 20 dB) on the MUUFL dataset. The SNR is formulated as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{E\left[\boldsymbol{y}^T\boldsymbol{y}\right]}{E\left[\boldsymbol{n}^T\boldsymbol{n}\right]} \tag{15}$$

where $E[\cdot]$ is the expectation of its content and $\boldsymbol{y}$ and $\boldsymbol{n}$ denote the clean pixel and the additive noise, respectively. The $ROC_{(D,F)}$ curves are displayed in Fig. 12, and it can be seen that the proposed TAEF still outperforms other competitors under three levels of noise. Moreover, 20 replicates for each level of noise are performed, and then, the mean value and the standard deviation of $AUC_{(D,F)}$ values are listed in Table V. It can be seen that our TAEF has significantly higher mean $AUC_{(D,F)}$ values and exhibits better noise robustness, which could be explained by the AE structure, TR's ability in characterizing local-to-global spatial information and EMLM (the physical knowledge) embedding. Besides, we conduct the noise experiments on the MUUFL dataset with fixed anomalies, and thus, the standard deviations listed in Table V are significantly smaller than those in the synthetic datasets with random anomalies in [15] and [57].

H. Summary

The superiority of the proposed TAEF has been demonstrated adequately by the above experimental results on the eight real datasets. Four main advantages can be briefly summarized as follows.

1) *Effectiveness:* The local-to-global spatial information between pixels is characterized precisely by the adopted TR encoder owing to its SA mechanism with the support

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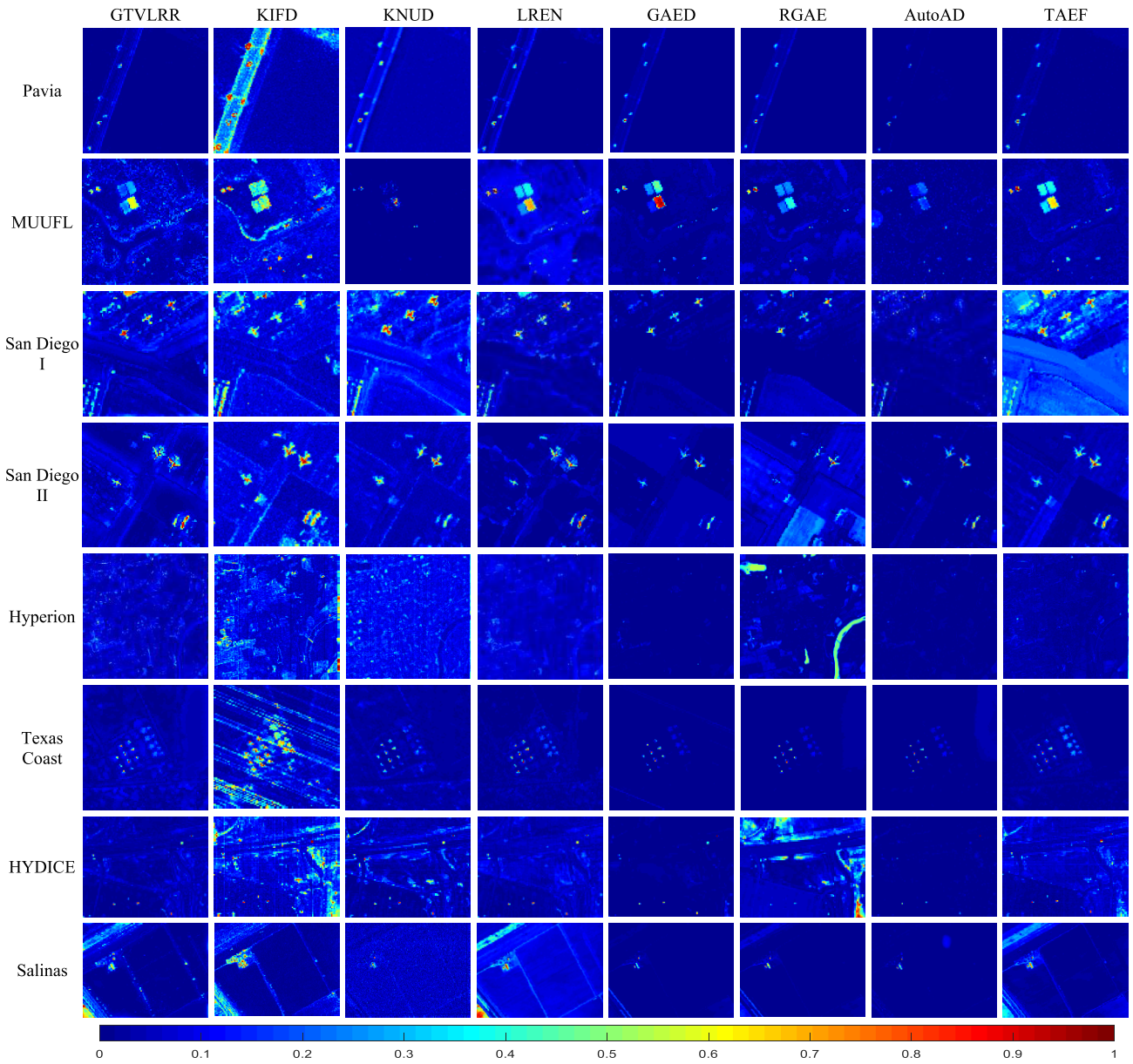

Fig. 10. Anomaly detection maps of all detectors on the eight real hyperspectral datasets.

TABLE IV

EXPERIMENTAL RESULTS OF ALL ANOMALY DETECTORS ON THE EIGHT REAL HYPERSPECTRAL DATASETS

Algorithm	GTVLRR	KIFD	KNUD	LREN	GAED	RGAE	AutoAD	TAEF
$AUC_{(D,F)}$ VALUES OF ALL ANOMALY DETECTORS ON THE EIGHT REAL HYPERSPECTRAL DATASETS.								
Pavia	0.9948	0.9851	0.9967	0.9959	0.9948	0.9927	0.9915	**0.9978**
MUUFL	0.9784	0.9863	0.9864	0.9948	0.9885	0.9832	0.9735	**0.9959**
San Diego I	0.9945	0.9861	0.9961	0.9929	0.9951	0.9940	0.9506	**0.9963**
San Diego II	0.9915	0.9953	0.9946	0.9923	0.9940	0.7629	0.9965	**0.9968**
Hyperion	0.9889	0.9925	0.9934	0.9908	0.9949	0.9408	0.9944	**0.9963**
Texas Coast	0.9927	0.9818	0.9853	0.9887	0.9959	0.9926	0.9673	**0.9980**
HYDICE	0.9930	**0.9952**	0.9925	0.9906	0.9808	0.7949	0.9932	0.9949
Salinas	0.9932	0.9832	0.9981	0.9964	0.9987	0.9987	0.9995	**0.9999**
Average	0.9909	0.9882	0.9930	0.9928	0.9928	0.9325	0.9833	**0.9970**
RUNNING TIMES OF ALL ANOMALY DETECTORS ON THE PAVIA DATASET.								
Training Time (s)	—	—	—	12.6901	13.9935	18.5429	8.7988	92.1263
Test Time (s)	82.2128	34.5528	122.7999	0.0317	0.0466	**0.0155**	0.0176	4.8488

of the utilized patch generation method. Moreover, the EMLM embedded decoder can accurately characterize the inherent high-order nonlinear mixing effects in HSIs. Consequently, the background reconstruction accuracy of the proposed TAEF is significantly better than those of the existing AE-based detectors. Besides, we adopt the

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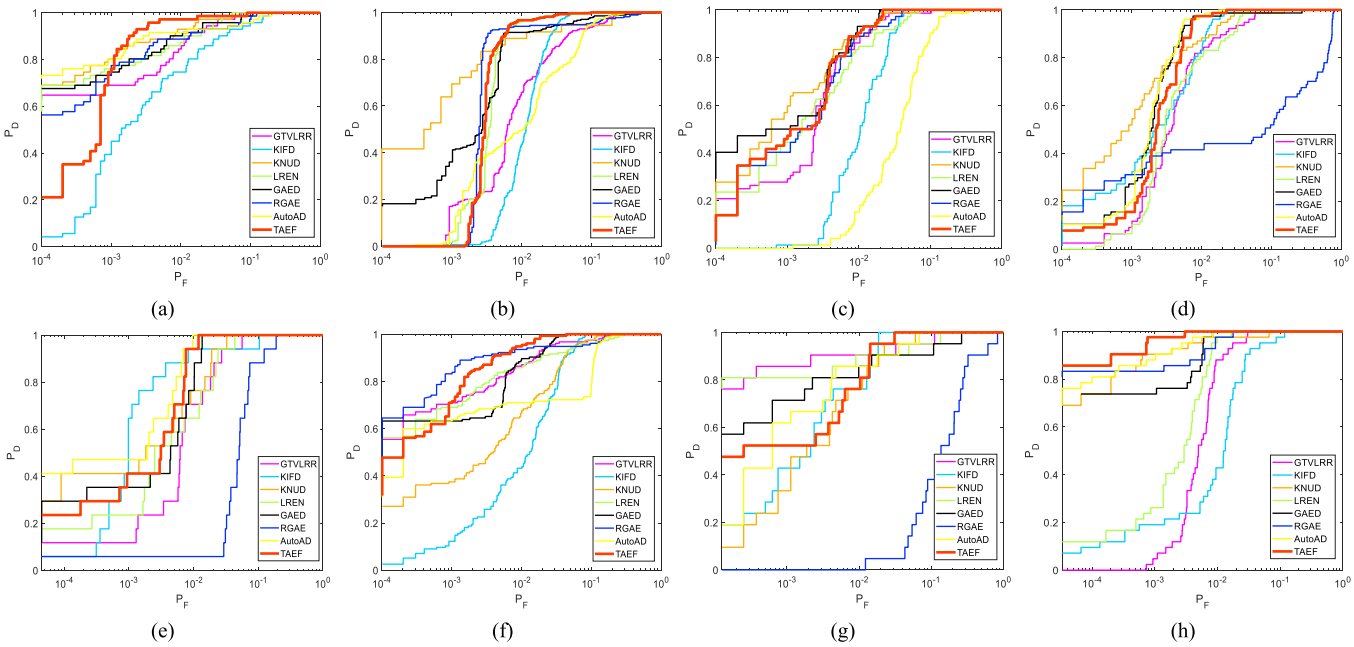

Fig. 11. $\mathrm{ROC}_{(D,F)}$ curves of all detectors on the eight real hyperspectral datasets. (a) Pavia. (b) MUUFL. (c) San Diego I. (d) San Diego II. (e) Hyperion. (f) Texas Coast. (g) HYDICE. (h) Salinas.

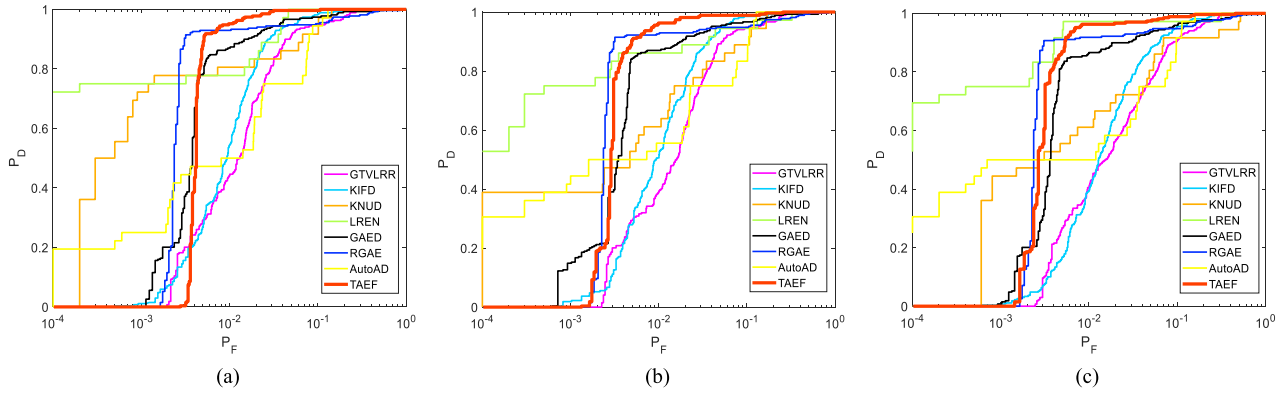

Fig. 12. $\mathrm{ROC}_{(D,F)}$ curves of all the detectors on the MUUFL dataset with different levels of zero-mean white Gaussian noise. (a) 30 dB. (b) 25 dB. (c) 20 dB.

TABLE V

$\mathrm{AUC}_{(D,F)}$ VALUES (MEAN $\pm$ STD) OF ALL ANOMALY DETECTORS ON THE MUUFL DATASETS WITH DIFFERENT LEVELS OF ZERO-MEAN WHITE GAUSSIAN NOISE (20 REPLICATES FOR EACH LEVEL OF NOISE)

Algorithm	GTVLRR	KIFD	KNUD	LREN	GAED	RGAE	AutoAD	TAEF
30dB	0.9748±0.0002	0.9837±0.0005	0.9847±0.0002	0.9933±0.0002	0.9876±0.0003	0.9827±0.0003	0.9725±0.0003	**0.9948±0.0001**
25dB	0.9712±0.0004	0.9816±0.0008	0.9757±0.0003	0.9918±0.0003	0.9851±0.0005	0.9819±0.0004	0.9706±0.0003	**0.9944±0.0002**
20dB	0.9622±0.0008	0.9732±0.0013	0.9504±0.0005	0.9904±0.0004	0.9812±0.0008	0.9707±0.0006	0.9680±0.0004	**0.9930±0.0004**

local-clustering method to handle the contamination of anomalies and reduce the false alarms caused by sparse backgrounds.

2) *High Efficiency:* Although the training time of the proposed TAEF is relatively high due to the iterative training on many epochs, the test time of the proposed TAEF is acceptable and relatively short in the fine-trained network, which shows the important practical application value of the proposed TAEF after training in advance.
3) *Robustness to Noise:* The MUUFL dataset corrupted by three levels of additive noise is used to assess the robustness of the proposed TAEF. The TR encoder is especially effective for eliminating noise because of its sufficient utilization of local-to-global spatial information. It can be seen from the experimental results that the proposed TAEF still obtains satisfactory detection results under different levels of noise.
4) *Simplicity of Choosing Parameter:* The loss function of the proposed TAEF only contains one loss item, and thus, there is no tradeoff parameter. Besides, the parameter experiments on the only involved parameter, the bandwidth of the LPF U, demonstrate that the anomaly detection result of the proposed TAEF is

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insensitive to the choices of U. Moreover, we fix the value of U and still obtain the satisfactory detection results on all the experiments.

It should be noted that the EMLM used in the proposed AE framework can be replaced by other mixing models. Therefore, the proposed TAEF is flexible enough to accommodate various hyperspectral datasets with different degrees of nonlinear mixing phenomenon. However, the replacement of the embedded model in the AE decoder is accompanied by the adjustment of the network structure, which may be inconvenient in practical applications. To solve this issue, the endmember-kernel theory [19] is worth researching so that any kind of nonlinear mixing models can be embedded into the AE without adjusting the network structure.

V. Conclusion

In this article, we have proposed a novel TAEF for nonlinear hyperspectral anomaly detection. Compared with the existing AE-based anomaly detectors that map and reconstruct each pixel separately, the adopted TR encoder can precisely characterize the local-to-global spatial information between pixels with the help of the utilized patch generating method (OWin), which benefits the reconstruction accuracy of background component. Moreover, the inherent nonlinear characteristics have been considered by the EMLM embedded decoder, which is more consistent with the complex light scattering effects in practical cases. It is worth noting that the EMLM can be replaced by other mixing models in the proposed AE framework. Besides, we have adopted the local-clustering method to adjust the proportions of the anomalies and the sparse backgrounds in the training dataset, which can significantly eliminate the contamination of anomalies and alleviate the false alarms. The noise experiments on the MUUFL dataset have demonstrated the robustness of the proposed TAEF. Besides, the ablation experiments and the parameter experiments on the two datasets and the real experiments on the eight datasets have verified the effectiveness of the proposed TAEF comprehensively.

To avoid adjusting the network structure in the application of various HSIs with different degrees of nonlinear mixing phenomenon, the endmember-kernel theory [19] is worth researching so that any kind of nonlinear mixing models can be embedded into the AE by simply replacing the kernel matrix.

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